

and clustering techniques to organise data. This helps to understand the structures and enable the discovery of patterns. These patterns are related to previously unknown and unnoticed scenarios such as identification of new users, service traffic behaviour and profiles, and are used to improve forecasting in network planning.

Self-aware, self-defending adaptive networks

A self-conscious, self-defending adaptive network system is a smart agent-based system that tracks network activity, information, and actions. Data focused on network activity is used to recognise and combat various forms of cyber threats.

“Artificial Intelligence is the new electricity.”
—**Andrew Ng**, founder and CEO, Landing AI

The architecture of the intelligent agent is so designed that it knows and recognises the level of threat faced by a node in a network. In order to identify and separate cyber security risks, the adaptive network architecture employs machine learning methods to track the network. The known security threats include ransomware, code hijackers, intrusion and illegal entry, piracy and unauthorised usage. This requires the self-awareness, self-protection and adaptation of all properties to any external or internal danger. This approach eliminates the possibility of zero-day attacks. This is because anomalous packet actions and information can be observed by the network. The adaptive network self-learning device learns through use and becomes wise with time.

The self-conscious, self-defending adaptive network system understands the actions and quality of each packet. These actions and content are used to assess whether the trend is predicted or anomalous. This adaptive and associative network senses increasing bytes in the system's partnership. It is able to identify trends in known threats, and can detect anomalous trends. The identified anomalous patterns may be associated with a zero-day attack, non-compliant network use or sabotage.

To demonstrate an efficient self-learning network, we are looking at the neocortex's individual neural network. A clever system, influenced scientifically, acts like the human brain, which learns autonomously at the moment of stimulus by recognising patterns. The adaptive network, like the human brain, stores each special byte sequence. The values, such as stimulus date, location of stimulation, pattern format, packet payload and addressing, are retained each time the pattern is observed. The data model stored has an n-dimensional representation of each pattern's semiotic value.

Intent based networks

Intent based networks are used to identify the business-related trends, and bridge the gap between the enterprise and IT. The advantages of purpose based networking are linked to the potential to intelligently simplify network management and orchestration. These reflect a solution set that is the confluence of key technologies such as machine learning, AI, software defined networking and IoT technologies.

The new network infrastructure system is called cognitive network management. Network operators use

“We are entering a new world. The technologies of machine learning, speech recognition, and natural language understanding are reaching a nexus of capability. The end result is that we'll soon have artificially intelligent assistants to help us in every aspect of our lives.”

—**Amy Stapleton**, co-founder, Chatables.

proven self-organising network systems to implement and run networks. Mixed with software defined networking and advanced analytics, artificial intelligence brings the automated, cognitive activity and management of the network to the next level. The combination of machine learning, software defined networks, and data analytics innovations will quickly lead to early and comprehensive performance in cellular networks. Future economic and social gains can be accomplished through self-awareness, self-configuration, self-optimisation, self-healing and self-protection in networks. 

By: Bhagvan Kommadi

'Quantica Computacao', India's first quantum computing startup, was founded by the author. He has 20 years of experience in product development and IT services.